SON CAO

Computer Science Student

CONTACT

+31 616411761

chson230905@gmail.com

hoagson2309

Son Cao

230 Zweringweg, 7545DA, Enschede

PROFILE

I'm a Computer Science student at Saxion University of Applied Sciences with a strong interest in Artificial Intelligence and machine learning. I enjoy learning by building and working on hands-on projects involving data, algorithms, and software development.
I'm proactive, adaptable, and always looking for ways to improve my skills. Currently, I'm seeking internships or part-time opportunities in AI, data, or software development where I can gain practical experience and contribute to real-world projects.

EDUCATION

Bachelor of Computer Science

Saxion University of Applied Science

2024 - 2028

SKILLS

- C / C++
- Python
- HTML / CSS
- C# / JavaScript
- Git
- Database / SQL
- Machine Learning
- Deep Learning
- Embedded systems (Rasberry Pi Pico, ESP32)

LANGUAGE

English

Vietnamese

PROJECTS

Embedded Temperature Sensor and Ambient Light Monitoring System (Raspberry Pi Pico - RP2040)

- Designed and implemented an embedded monitoring system on a Raspberry Pi Pico (RP2040) using I2C and SPI to integrate LM75B temperature sensor (0.125°C, 11-bit) and APDS-9306 ambient light sensor (20-bit), including register-level configuration, data processing, and interrupt-based threshold detection.
- Developed a real-time system using GPIO interrupts for button control, temperature (OS) and light (INT) alerts, and SPI-controlled LED output via MCP23S08, implementing debouncing, bitwise data handling, and serial debugging for validation.

NVIDIA (NVDA) Stock Price Prediction

- Built an end-to-end machine learning pipeline (2022–2026 data) to predict next-day NVIDIA (NVDA) stock returns using Alpaca market data, including feature engineering with technical indicators (momentum, trend, volatility) and a Random Forest model evaluated using MAE.
- Analysed and visualised model performance, highlighting key challenges in financial time-series forecasting such as error accumulation and the trade-off between prediction accuracy and stability.

Real-Time Hand Gesture Recognition (CNN + Temporal Convolutions)

- Built a real-time gesture recognition system using CNN and temporal Conv1D layers to classify hand gestures (thumbs up, thumbs down, stop) from video sequences. Created a data pipeline with OpenCV and trained the model using TensorFlow/Keras, with real-time webcam prediction.
- Achieved good validation results on a custom dataset and successfully ran real-time gesture classification on a standard laptop, combining both spatial and temporal feature learning for video input.